Notes (Week 3 Wednesday)

These are the notes I wrote during the lecture.

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The number of 2-tuples (pairs)
  (x,y) ∈ S × T

  |S| * |T|

Alternative question:
  how many 0-tuples are there?

  () ∈ Π_{}. foo

  A: 1

  Π_i∈{1,2,3}. i = 1*2*3

  Π_i∈(S ∪ T). i     (supposing S,T disjoint)

  Π_i∈(S ∪ T). i = Π_i∈S. i * Π_i∈T. i

  ^ in order for this law to work when T=∅,
    we need the product of no numbers to be 1.


|V| = 5

|V^0| + |V| + |V^2| + |V^3|

1 + 5 + 5*5 + 5^3 = 156

Step1: translate A*VA* into English
  V means "a vowel"
  A means "a letter"
  A* means "0 or more letters"
Therefore, A*VA* means:
  - 0 more letters (any letter)
  - followed by one vowel
  - followed by 0 or more letters

C*VC* this becomes:
  - 0 more consonants
  - followed by one vowel
  - followed by 0 or more consonants

V(CV)* the words that:
  - start with a vowel,
  - followed by 0 more consonant-vowel pairs
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V+(CV+)∗

  - V+:  one or vowels
  - CV+  a consonant, followed by one or more vowels

Words comprised of:
  - one or more vowels, followed by
  - zero or more groups of one consonant followed by
    at least one vowel


 sum(n) = sum2(n) for n ∈ ℕ

By basic induction, it suffices
to prove:

(B) sum(0) = sum2(0)

    sum(0) = 0
    = 0∗(0+1)/2 = sum2(0)

(I) If we assume
      sum(n) = sum2(n) (IH)
        "induction hypothesis"
    then it follows that
      sum(n+1) = sum2(n+1)


      sum(n+1) =
      (n+1) + sum(n) = (by IH)
      (n+1) + sum2(n) =
      (n+1) + n∗(n+1)/2 =
      (a bunch of algebra) =
      sum2(n+1)

Assuming we've accepted that
  basic induction is valid,
you can prove that induction from m upwards
is valid using basic induction.

Assuming
  (i) P(m),
  and (ii) ∀x ≥ m. P(k) → P(k+1)

  I would like to prove: ∀n ≥ m. (P(n))

  Observation: ∀n ≥ m. (P(n))  ≡  ∀k. (P(m+k))

  Then, I can prove ∀k. (P(m+k))
  by basic induction on k.
  For that proof by basic induction, I get
  two proof obligations:
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   (B) P(m+0) <-- immediate from (i)
   (I) P(m+k) → P(m+k+1) <-- follows from (ii),
                             by reasoning similar
                             to the "observation"

fib(0) = 0
fib(1) = 1
fib(n) = fib(n-1) + fib(n-2)

Every 4th Fibonacci number is divisible by 3,
by 4-step induction.

  (B) fib(4) = fib(3) + fib(2)
             = fib(1) + 2*fib(2)
             = 3*fib(1) + 2*fib(0)
             = 3
  (I) Assume 3 | fib(n)

      Prove 3 | fib(n+4)

      fib(n+4) = fib(n+3) + fib(n+2)
               = 2*fib(n+2) + fib(n+1)
               = 3*fib(n+1) + 2*fib(n)

      ^fib(n+4) is the sum of two numbers that
          are divisible by 3, and therefore,
          it's divisible by 3



Strong induction (alternative presentation)

To prove ∀n ∈ ℕ. P(n) holds, it suffices to prove

(B+I)  For all n, assuming
           ∀m < n. P(m)
       then P(n) holds

 ^ that's a base case when n = 0,
   it's an inductive case otherwise

An inductive case is any case with at least
one inductive hypothesis.

The more general principle is called
  Noetherian induction (Emmy Noether)

Well-founded strict poset

Strict poset -- strict partially ordered set

A strict poset (<,S) such that < ⊆ S × T
such that:
  < is antireflexive
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       antisymmetric
    and transitive

Well-founded: has no infinitely descending chains.
 Chain is a sequence of elements of S linked by <
 Descending chain has a largest element.

 (<,ℕ) is well-founded

    I'm stuck :(  0 < 1 < 4 < 5 < 7

 (<,ℤ) is *not* well-founded

            ... < -4 < -3 < 0 < 1 < 4 < 5 < 7

 (≤,ℕ) is this well-founded?

         .. ≤ 0 ≤ 0 ≤ 0

Q: Does that mean if something is well-founded
   it has to be strict?
A: yes

Alternative presenation of well-founded induction:
  Suppose (<,S) is a well-founded strict poset.
  In order to prove ∀s ∈ S. P(s), it suffices to prove:

   For all s, if
      ∀t<s. P(t)
   then P(s) holds

*Every* induction principle (yes, every)
  are special cases of Noetherian (aka well-founded)
  induction.

Q: why is well-foundedness important?
A: if a poset is well-founded, intuitively,
   the base case is always reachable.
   or, in programming terms, your thing terminates

Q: What's the connection between induction on
   Σ* and well-founded induction?

A: In the same way that induction on natural
   numbers is well-founded induction
   with (<,ℕ),

   induction on Σ* is well-founded induction
   with

     (<,Σ*)

   <  is "suffix inclusion"
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   w < w'   iff w is a suffix of w'
   a word is a suffix of another if it
   occurs at the end of it

    "lo"  is a suffix of "hello"

Prove this by induction on w:
  length(wv) = length(w) + length(v)

(B) length(λv) = length(λ) + length(v)

    length(λv)
    = (by def of concatenation)
    length(v)
    = 0 + length(v)
    = length(λ) + length(v)

(I) assuming
     (IH) length(wv) = length(w) + length(v)
    prove
     length(awv) = length(aw) + length(v)

     length(awv)
     = (by def of length)
     1 + length(wv)
     = (by IH)
     1 + length(w) + length(v)
     = (by def of length backwards)
     length(aw) + length(v)



odd(0) = false
even(0) = true
odd(n) = even(n-1)
even(n) = odd(n-1)

^ we could collapse this into a single function
  as follows:

 oddeven(0,b) = b
 oddeven(n,b) = oddeven(n-1,¬b)

These should satisfy:
 oddeven(n,false) = odd(n)
 oddeven(n,true) = even(n)



We've seen several inductive proof techniques for ℕs.
They're all equally strong.

  You can prove the exact same propositions using
   basic, or complete, or whatever induction,
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   but sometimes, one of them is more convenient.

Every well-founded poset has an induction principle.

We've seen how you can use this to derive (structural)
induction on words, and induction on ℕs.



(B)  E denotes the empty tree (leaf node)
(I)  N(l,v,r)  denotes a non-leaf node with
              left subtree l, right subtree r and
              value v.

  ^ that's an inductive definition of a set
Therefore, we can prove facts about trees using
tree induction!

   To prove P(t),
   it suffices to prove
   (B) P(E)
   (I) If P(l) and P(r) then P(N(l,v,r))
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